

Viscous Damping Device for Tremor Reduction in Parkinson's Patients

Senior Design I – Spring 2025

Kansas State University

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Introduction

Parkinson's disease is a neurodegenerative disease which primarily affects movements often causing symptoms such as tremors. Tremors are moments of involuntary shaking, most commonly appearing in the hands. The tremors can reduce independence, self-esteem and overall quality of life. Currently about one million people are affected by Parkinson's disease in the United States, interfering with their day-to-day life's.

To address this, our team has developed a wearable viscous damping device that is designed to reduce tremors in the wrist through fluid resistance.

This report presents our device design, safety analysis, environmental sustainability, patent search, economic feasibility and social impacts. Our primary objective with this device is to develop it as a cost effective and functional solution that can enhance the motor control and support mobility for Parkinson's patients.

Public Safety Analysis

Covered Safety Topics

Our design addresses many safety factors:

- **Material Safety:** Majority of the device will be injection molded ensuring high precision, consistency, material strength, smooth surfaces, durability, and materials that withstand disinfection.
- **Mechanical Safety:** The device has no pinch points or sharp edges that may affect the user during wear. The damping involved is passive and cannot over restrict the user's movement during day-to-day activities.
- **Electrical Safety:** Due to the device incorporating sensor, control systems and battery powered components we want to ensure the users protection. Electrical safety is important to ensure the device is safe, dependable and legally compliant.

Follow-Up Testing Required

To ensure our device is in full compliance with safety standards, the following testing phases are planned:

- Ergonomic testing for the wearability and comfort of the device on the user's wrist.
- Long-term durability testing of the internal damping fluid.
- Electrical safety evaluation for the sensor and power systems within the device.

Flowchart for Safety Testing Processes

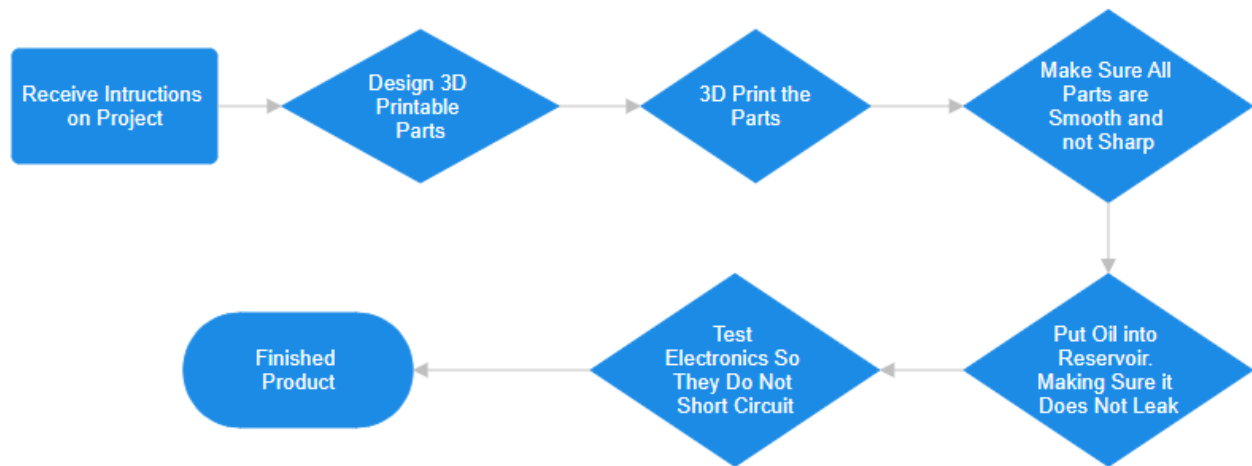


Figure 1 – Safety Testing Process Flowchart

Relevant Safety Standards

For our wearable wrist damping device, our product must comply with a range of safety and medical device standards. Below are listed some of the applicable standards and how our design aligns with each:

1. ISO 13485: 2016 – Medical Devices: Quality Management Systems

This standard outlines specific requirements to ensure medical devices meet customer and regulatory demands for safety and efficacy. During our time, we followed structured design control processes which included documentation, traceability and risk management. Our team-maintained material sourcing and part inspection to ensure quality and reproducibility during manufacturing of our device.

2. ISO 14971: 2019 – Medical Devices: Application of Risk Management to Medical Devices

This document describes the process of the risks associated with a medical device, including the identification, evaluation and control throughout the product lifecycle. To comply with this standard, a risk analysis was conducted to our device to assess any potential hazards such as fluid leaks, skin irritation, and electrical failure. We implemented risk controls such as dealing enclosures, using skin safe materials throughout and including low-voltage electronics.

3. IEC 62366: 2015 – Usability Engineering to Medical Devices

This standard specifies a process for manufacturers to analyze, specify, design and evaluate the usability of medical devices with a focus on safety. Our device incorporated user centered design principles. This was done by testing the usability, ensuring strap accessibility by using simple straps, and making adjustments for users with limited dexterity by avoiding complex interfaces or buttons.

By aligning our devices design with the recognized safety standards such as ISO 13485, ISO 14971, and IEC 62366, we have ensured that our wearable wrist damping device meets the requirements for medical safety, reliability and user comfort. These standards helped guide our decisions in materials, risks, usability and electrical design. As a result, our product prioritizes health and safety of individuals with Parkinson's disease, as well as lays a strong foundation for future approval and real-world implementation. Continued testing and refinement on our product will validate our commitment to delivering a safe, accessible and effective solution for Parkinson's tremors.

Environmental Sustainability of Design and Production

Sustainability Areas Analyzed

There are many areas of our project that will be sustainable. One example, though very unfortunate, is the fact that there are over 1.1 million cases of Parkinson's Disease in the United States alone. While not all the cases involve wrist tremors, the vast majority of people who have Parkinson's deal with some type of wrist tremor. This allows for our design to be useful for years to come until a more permanent solution for Parkinson's is found.

Design Features Supporting Sustainability

To ensure the sustainability of our design we made all our custom designed parts injection moldable to enable a longer life span of our units. Injection molding allows for parts to be easily created for a low price per unit and each injection molded part would be stronger and able to handle more stress than an average 3D printed part.

Patent Search and Analysis

Patent Research

To ensure our viscous damping device is both original and viable, we conducted a comprehensive patent search. Our goal was to evaluate the current intellectual property related to wearable tremor reducing technology and determine if our devices design presents a patentable

innovation. The patent databases we used were Google Patents and United States Patent and Trademark Office (USPTO). We used keywords such as “viscous damping device,” “fluid damping wrist device,” “Parkinson’s stabilization system,” and “silicone watch band.” Listed below are some of the patents we found relevant to the design of our device:

1. U.S. Patent No. 4689449A

This patent covers a wrist support with shock absorbers. The mechanism reduces repetitive stress or vibrations. The patent is relevant to our design as it includes a damping element in the wrist worn format which is similar to our design. Since this technology is more related to repetitive motion injuries than medical tremor suppression it does not impact our design because our fluid based system distinguishes it in function and application.

2. E.P. Patent No. 3562443B1

This European patent is for an orthotic device which is designed to passively reduce tremors through an adjustable mechanical component and dynamic restriction. This patent is highly relevant to our design as it includes modularity and passive tremor control in a wearable format. While this design presents a closer match to our design than other patents, our use of viscous fluid damping and silicone wrist band makes our device design unique. This patent is still something to monitor in future iterations or international expansion.

Visual Summary of Patents

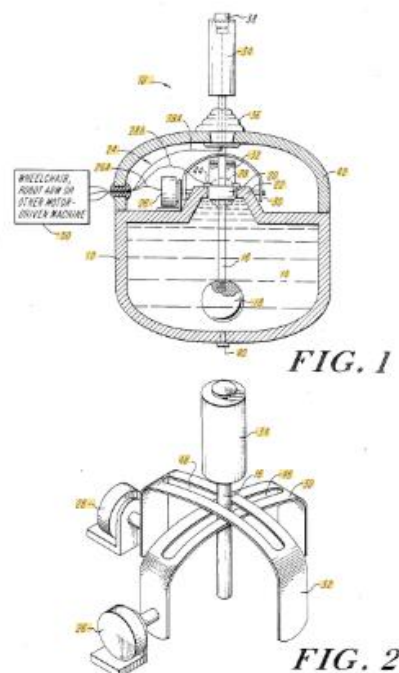


Figure 2 - Tremor Suppressing Hand Controls

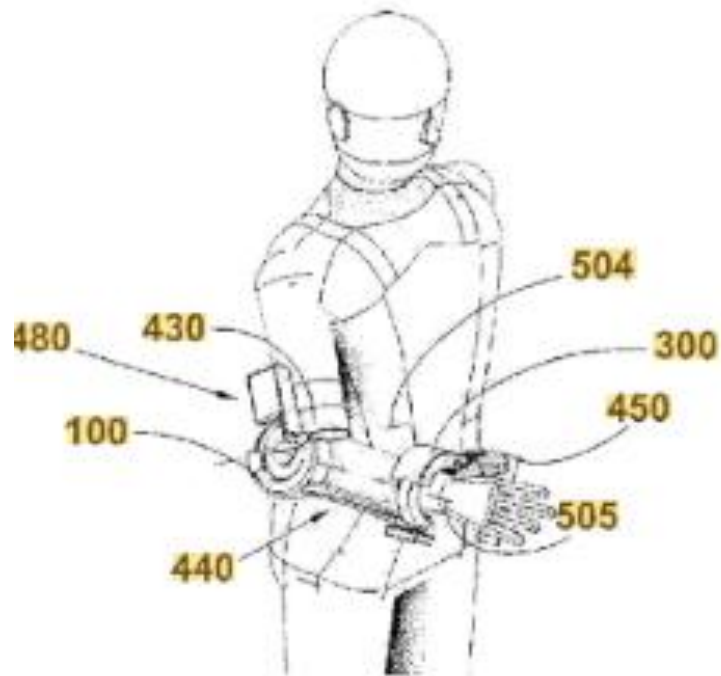


Figure 3 - Fluid Dampening Exoskeleton

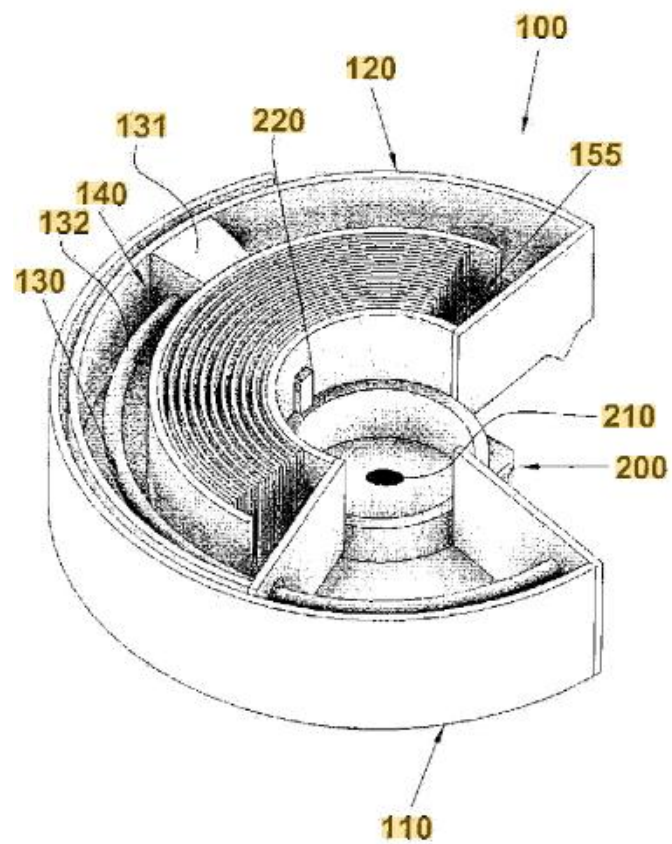


Figure 4 - Electromagnetic Reservoir

Patentability and Market Impact

Our analysis after patent research indicates that there are no current patents directly combining ergonomic design, wearable comfort and viscous fluid damping in a low powered wrist device for Parkinson's tremor reduction. While there are other systems that exist for wrist support or electronic stabilization, our hybrid design stands apart in the following ways:

- Our innovation lies within our viscous fluid damping mechanism that is integrated into a lightweight and wearable reservoir.
- The use of a silicone band on the wrist has enhanced the comfort and usability but does not duplicate other patented mechanisms.
- Our device's design is simpler, more affordable and accessible to a broader user base while including Parkinson's disease patients.

Our current design of our fluid damping device appears to be distinct in functionality, application and structure when compared to other existing patents. While the design shows some similarities in overall intent such as tremor reduction and wrist worn form, our unique use of passive fluid damping creates a meaningful differentiation. Our design is likely to become patentable with furthest testing and proper technical distinction which could benefit from a utility patent filing to protect our mechanism and integration approach.

Economic Analysis and Competition

Cost Analysis

- Production Cost per Unit: \$17.16
- Total Cost per 1,000 Units: \$17,160.00
- Injection Tooling Cost: \$52,000.00 (One time cost)
- Retail Price: \$70.00
- Estimated Units Sold in 2-3 Years: 1000 units

Breakeven Point

To determine the breakeven point for our product, we calculated how many units needed to be sold to recover from the initial tooling cost, assuming all the other production costs were covered by the markup on each unit.

- Profit per Unit = Retail Price – Cost per Unit
- Profit per Unit = \$70.00 - \$17.16
- Profit per Unit = \$52.84
- Breakeven Units = Tooling Cost / Profit per Unit
- Breakeven Units = \$52,000.00 / \$52.84
- Breakeven Units = 984 Units

We calculated the breakeven point to be when 984 units were sold. In the long run, the projected 1,000-unit sales in 2-3 years would begin generating profit after recovering the tooling investment.

Market Opportunity

Our product has been positioned as a mid-range solution in the market for assistive tremor reduction devices for Parkinson's patients. Compared to the active products on the market which range in the hundreds to thousands of dollars, our device offers a low cost, minimal maintenance, and greater wearability. At a \$70.00 retail price, it is competitively priced for an accessibility product while still providing value and meeting medical expectations.

Competitive Landscape

Table 1 - Solution Comparison

Product	Type	Price	Pros	Cons
Our Device	Passive	\$70.00	• Affordable • Ease of use • Wearable	• New to market • Needs further testing
GyroGear	Active	\$5,899.00	• High tech • Ambidextrous • Multiple sizes	• Expensive • Limited access
Stediwear	Passive	\$599.00	• Instant stability • Lightweight • Battery free	• Hand/Wrist specific • Limited adjustability
Readi-Stedi Orthotic Glove	Passive	\$600.00	• Customizable • Comfortability • Non-invasive	• Requires physician order • Insurance verification

This competitive analysis demonstrates a potential market opportunity for our device as it is affordable, easy to use, wearable for everyday use and is an accessible solution to patients with Parkinson's disease. While some of the current products on the market like GyroGear and Stediwear offer advanced or customizable features, their high costs and limited accessibility creates a barrier for user purchase. Our viscous damping device fills a gap in the market by offering passive tremor damping at a fraction of the cost, with minimal setup and high user comfort. Although further testing is needed to validate performances across diverse users, our device's design offers a strong balance of functionality, cost efficiency and manufacturability. This positions it well for entry into both the medical and consumer markets.

Social Impacts of the Design

The primary social impact of our design is an improvement in independence and quality of life for individuals with Parkinson's tremors. By allowing better control during daily activities, the device may also reduce the stigma associated with the visible tremors from the individuals. Additionally, the low cost and slim design make it accessible and less clinical in appearance, encouraging the prolonged use of the device.

Conclusion

Our viscous damping wrist device represents a meaningful step towards improving motor controls for individual affected by Parkinson's tremors. It is safe, sustainable, economically feasible and socially beneficial. With further testing and refinements on the device, this product has strong potential for marketability, clinical use and significant impact in assistive medical technology.

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